

ArTiSan: Archetype-Aware TimeSformer for Fast Glaucoma Progression

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Abstract. Glaucoma is a leading cause of irreversible vision loss, and accurate detection of this disease’s progression from longitudinal visual field (VF) data remains a longstanding challenge. Glaucoma progression often manifests through characteristic patterns of VF loss. Archetypal Analysis represents each VF as a convex combination of learned archetypal defect patterns. In prior work, these archetypes were shown to correspond to clinically interpretable patterns of VF loss. In this study, we leverage these representations as complementary information for glaucoma progression detection. Specifically, archetypal weight vectors are incorporated as additional tokens within a TimeSformer-based spatiotemporal attention framework. Experiments on the UWHVF and LEI-SFD datasets show improvements over the baseline model in several classification metrics, particularly at moderate-to-fast progression thresholds. These findings provide preliminary evidence that archetypal representations offer complementary pattern-level information to spatiotemporal VF features, with the clearest benefit observed in the fast-progressing cases.

Keywords: Glaucoma progression · Visual Field · Archetypal Analysis

1 Introduction

Glaucoma is a progressive optic neuropathy that causes structural changes to the optic nerve head, resulting in irreversible vision loss. While vision loss from glaucoma cannot be restored, its progression can be slowed in most cases, making early detection and accurate monitoring essential for treatment decisions and the preservation of remaining vision [11]. The standard clinical tool for assessing functional progression is the Humphrey Visual Field (HVF) test, a form of static automated perimetry that maps the sensitivity of the visual field (VF), as shown in Fig. 1a. The HVF test also computes Total Deviation (TD) values, which represent the difference between a patient’s measured light sensitivity and the

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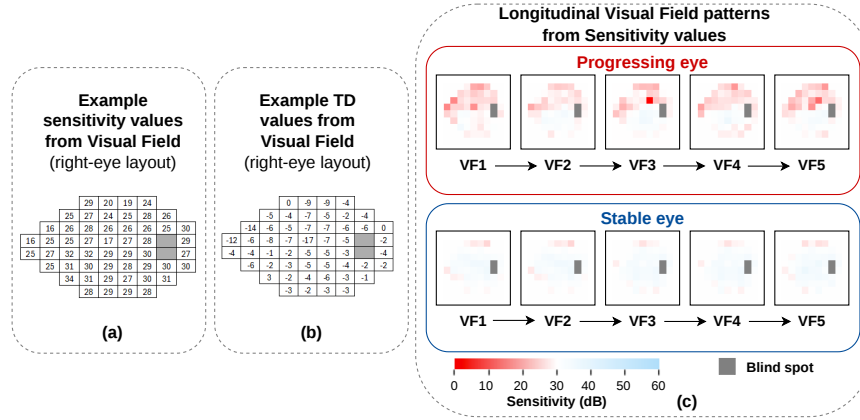


Fig. 1. Example visual fields (VFs) from the UWHVF dataset [9], acquired using the Humphrey Visual Field (HVF) 24-2 test pattern with 54 test locations, including two blind-spot locations. (a) Sensitivity values range from 0 to 50 dB, with lower values indicating reduced visual sensitivity. (b) Total deviation (TD) values represent the difference from age-corrected normal sensitivity, where increasingly negative values indicate greater VF loss. (c) VF sensitivity is visualized using a color-coded gradient of red, white, and light blue [10]. 5 examples of progressing VF from patient A’s right eye and 5 examples of stable VF from patient B’s right eye are illustrated in sequence.

age-matched normal sensitivity as shown in Fig. 1b. Other than sensitivity and TD, HVF also prints out global indices, Mean Deviation (MD) and Pattern Standard Deviation (PSD) that summarise the overall level and localisation of VF loss.

VF measurements are inherently noisy, affected by patient attention, fatigue, and test-retest variability [5], making reliable detection of genuine progression from longitudinal series a persistent clinical challenge, as illustrated in Fig. 1c. One way to reduce the effect of noise is to represent each VF in terms of its underlying spatial pattern rather than its raw pointwise sensitivity values [3]. Archetypal Analysis (AA) provides such a representation by decomposing each VF into a convex combination of a small set of canonical loss prototypes called archetypes [10].

Conventional VF progression analysis methods commonly rely on global and pointwise trend-based indices [12]. While these methods are clinically interpretable, they typically require many follow-up examinations before progression can be detected with confidence. Recent deep learning approaches have investigated whether early VF sequences can predict later functional deterioration, including LSTM-based models [1], Convolutional LSTM models [1,13], and transformer-based architectures [1,2,15]. Tian et al. [15] proposed the Glaucoma Progression TimeSformer (GPTS), which treats the first three longitudinal VFs as a short video-like sequence and classifies future progression rate using divided space-time attention.

Although GPTS model can learn spatial and temporal dependencies from TD data [15], it does not explicitly encode recurring glaucomatous patterns of visual-field loss. Glaucomatous VF loss follows anatomically determined spatial patterns, such as nasal steps, paracentral defects, and partial arcuate defects [7], which are not directly represented in pixel-level patch tokens. To capture pattern-level structure in a compact and interpretable form, AA provides a complementary representation of VF [10].

Thakur et al. [14] demonstrated that convex representations of VF loss derived from deep archetypal analysis can classify glaucomatous from non-glaucomatous VFs, capturing preclinical signals inaccessible to global summary statistics such as MD and PSD. Wang et al. [17] further showed that archetypal weights over time can detect VF progression, outperforming conventional rule-based methods in clinical validation. To our knowledge, no prior work on glaucoma progression detection has incorporated AA-derived features within a deep learning pipeline.

We hypothesise that archetypal weight vectors can complement the spatiotemporal representations learned by TimeSformer [15] by providing pattern-level information relevant to progression detection. To test this, we propose the Archetype-aware TimeSformer (ArTiSan), with the following contributions:

1. An architecture incorporating archetypal weight vectors from 18 generalised VF archetypes as additional input tokens within a spatiotemporal transformer for glaucoma progression detection.
2. An evaluation against a strong baseline [15] on two longitudinal VF datasets, showing improvements in several classification metrics at the severe progression threshold (-1.5 dB/year).
3. Preliminary evidence that pattern-level AA features can support glaucoma progression detection within a spatiotemporal deep-learning pipeline.

2 Method

We formulate glaucoma progression detection as a binary classification task from early longitudinal VF data, following Tian et al. [15]. For each eye, the first three VF visits are used as model input: $\mathcal{V}_{\text{input}} = \{VF_1, VF_2, VF_3\}$. The target label is determined from the future follow-up visits, $\mathcal{V}_{\text{future}} = \{VF_4, VF_5, \dots, VF_n\}$. Given the first three VFs, the model predicts the progression label:

$$y = f(\mathcal{V}_{\text{input}}), \quad y \in \{0, 1\} \quad (1)$$

Where f represents the predictive function, mapping early VF trends to a binary state ($y = 1$ indicating progression and $y = 0$ indicating stability). The early VF visits are used as input and later VF visits define the progression target.

2.1 Preprocessing and Label Generation

The VF has 54 pointwise TD values. Each VF is represented as a 8×9 matrix of TD values following right-eye layout (Fig. 1a). The matrix is padded to enable 3×3 patch tokenisation, i.e., $V_t \in \mathbb{R}^{8 \times 9} \rightarrow V'_t \in \mathbb{R}^{9 \times 9}$.

The additional locations required to form a 9×9 matrix are zero-padded. Each preprocessed VF is then replicated across three channels to match the image-like input expected by TimeSformer, yielding $X_t \in \mathbb{R}^{3 \times 9 \times 9}$. The first three visits are stacked to form the model input, where the dimensions correspond to frames, channels, height, and width:

$$X \in \mathbb{R}^{3 \times 3 \times 9 \times 9} \quad (2)$$

Progression labels are generated from future visits only, beginning at VF_4 . For each future visit, Mean Deviation (MD) is approximated by averaging valid TD values, and Age represents the patient's age (in years) at the time of examination. Consecutive MD rates are then calculated as

$$r_t = \frac{MD_{t+1} - MD_t}{Age_{t+1} - Age_t}, \quad t = 4, 5, \dots, n-1. \quad (3)$$

The progression threshold is defined by the MD rate, with three values considered: $\tau \in \{-0.5, -1.0, -1.5\}$ dB/year. These thresholds represent different progression rates, with more negative values corresponding to faster VF deterioration. A rate of change below -1.0 dB/year is generally regarded as fast progression as measured by the MD slope [6].

An eye is labelled as progressing if the minimum future MD rate is below the threshold τ :

$$y = \begin{cases} 1, & \text{if } \min_t(r_t) < \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

2.2 Archetype Weight Extraction

In this study, we use predefined VF archetypes from the generalized VF pattern discovery proposed by Mutiawani et al. [10]. The study used Archetypal Analysis to identify generalized VF patterns and reported that 18 distinct patterns were obtained from UWHVF and consistently observed in eFOVID datasets[8]. In this work, these 18 UWHVF-derived archetypes are used as fixed reference patterns.

For each input VF, archetypal weights are computed relative to these fixed archetypes. Let $a_i \in \mathbb{R}^{18}$ denote the archetypal weight vector for sample i , where each element represents the contribution of one predefined VF archetype. The weights satisfy the convexity constraints:

$$a_{i,k} \geq 0, \quad \sum_{k=1}^{18} a_{i,k} = 1. \quad (5)$$

2.3 Model Architectures

We compare the proposed ArTiSan model with the baseline Glaucoma Progression TimeSformer (GPTS) to assess the impact of incorporating archetypal representations on glaucoma progression detection. Tian et al. [15] reported GPTS better performance over Vision Transformer and CNN (Convolutional Neural Network) models for longitudinal VF progression.

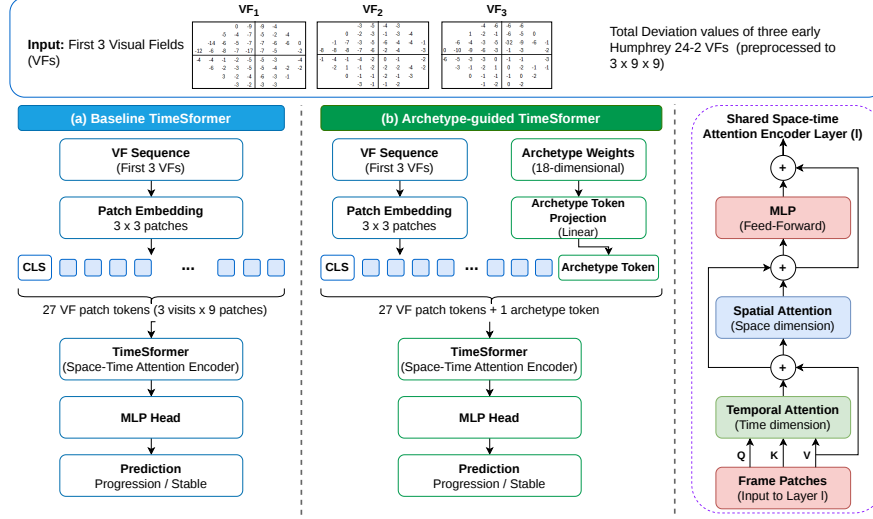


Fig. 2. Comparison of the baseline GPTS [15] and the proposed Archetype-aware model (ArTiSan). GPTS processes the first three visual fields as patch tokens. ArTiSan additionally appends an archetype token. The rightmost shared space-time attention encoder layer figure is the inner process of TimeSformer in both GPTS and ArTiSan.

Baseline GPTS: The baseline model follows the Glaucoma Progression TimeSformer formulation as proposed by Tian et al. [15]. The baseline model receives the three VFs in as described in Equ. (2). Each 9×9 VF frame is divided into 3×3 patches, producing 9 patch tokens per frame and 27 VF patch tokens across three visits. These tokens, together with a classification token, are processed by a space-time attention encoder. The final classification-token representation is passed to an MLP classification head, which outputs the progression logit $z = f_{\text{GPTS}}(X)$. The probability of progression is then computed as $\hat{p} = \sigma(z)$, where σ denotes the sigmoid function.

ArTiSan: The proposed Archetype-aware model, ArTiSan, uses the same VF patch-token sequence as GPTS, while additionally incorporating an 18-dimensional archetypal weight vector. The archetypal weight vector is projected into the transformer embedding space as $e_{\text{arch}} = W_1 a_i$, where only the first 18 UWHVF-derived archetypes are used as fixed reference patterns. This projected embedding is treated as an additional archetype token and appended to the input token sequence:

$$[\text{CLS}, \text{VF patch tokens}, \text{Archetype token}].$$

The archetypal weight vector a_i is obtained using a sensitivity-derived archetypal basis learned from previously established VF patterns [10]. In this work,

Table 1. Class distribution across datasets and MD-rate thresholds.

Dataset	Total Data	Threshold	Progressing	Stable
UWHVF	2991 eyes	-0.5	2665	326
		-1	2200	791
		-1.5	1769	1222
LEI-SFD	324 eyes	-0.5	322	2
		-1	302	22
		-1.5	231	93

these coefficients are computed from sensitivity-based VFs and used as a global structural descriptor (Subsection 2.2). This descriptor is then injected into a transformer operating on TD-based patch tokens. The combined sequence is processed by the same TimeSformer encoder, enabling joint attention over spatiotemporal VF patches and archetype-based global pattern information. The final classification token is passed to an MLP head to predict glaucoma progression, as illustrated in Fig 2.

3 Experimental Settings

3.1 Datasets

Two longitudinal VF datasets are used in this study. The University of Washington Humphrey Visual Field (UWHVF) dataset [9] is used for model development and internal evaluation. The Lions Eye Institute Structure-Function Dataset (LEI-SFD) [16] is used as an external test cohort to assess cross-dataset generalisation. Both datasets are converted to a right-eye layout for consistent analysis. Below settings are used for evaluation:

1. **Internal UWHVF evaluation:** The dataset is split into an 80 : 20 train-test ratio. The training set is evaluated using 5-fold Stratified Cross-Validation for model validation. The final evaluation on the 20% held-out test set is performed using a 5-model ensemble.
2. **External LEI-SFD evaluation:** The models are trained on UWHVF and evaluated directly on LEI-SFD, with no LEI-SFD samples used for training.

3.2 Class Distribution

Eyes with fewer than four serial VFs were excluded from both datasets. Table 1 shows the progression label distribution across datasets and MD-rate thresholds. As expected, lower thresholds reduce the number of progressing eyes. Both UWHVF and LEI-SFD datasets remain progression-dominant at thresholds -0.5 and -1.0 dB/year but become less imbalanced at -1.5 dB/year threshold.

Table 2. Performance comparison between GPTS and ArTiSan across MD-rate thresholds for the UWHVF dataset. Bold indicates the better result for each metric and threshold.

Metrics	-0.5 dB/year		-1 dB/year		-1.5 dB/year	
	GPTS	ArTiSan	GPTS	ArTiSan	GPTS	ArTiSan
Accuracy	0.892	0.892	0.743	0.756	0.681	0.691
Precision	0.892	0.892	0.760	0.788	0.717	0.739
Recall	1.000	1.000	0.952	0.916	0.760	0.737
F1-score	0.943	0.943	0.845	0.847	0.738	0.738
AUC	0.721	0.725	0.729	0.732	0.730	0.734

3.3 Implementation Details

Both models were implemented in PyTorch and trained as binary classifiers using binary cross-entropy with logits loss (implemented as *BCEWithLogitsLoss*). They share the same TimeSformer backbone, configured with three input VF frames, three duplicated channels per frame, 9×9 spatial resolution, 3×3 patch tokenisation, an embedding dimension of 512, 12 encoder layers, 8 attention heads, and a head dimension of 64. Models were optimised using SGD with a fixed learning rate of 0.1 and a mini-batch size of 5. Stratified 5-fold cross-validation was applied within the UWHVF training set, with training capped at 100 epochs and early stopping based on validation loss as the monitored criterion to prevent overfitting. Binary progression labels were obtained by thresholding the sigmoid output at 0.5.

4 Results and Discussion

We summarize the internal evaluation results on the UWHVF dataset in Table 2. While the performance of GPTS and ArTiSan is similar at the -0.5 dB/year threshold, ArTiSan achieves modest gains in several metrics at the -1.0 and -1.5 dB/year thresholds, particularly in accuracy, precision, and AUC. These results suggest that the addition of archetypal information produces a modest shift towards more precise, rather than more sensitive, progression predictions at the stricter progression-rate thresholds.

Table 3 reports the cross-dataset evaluation results on the LEI-SFD dataset. At the -0.5 dB/year threshold, both models achieve almost identical thresholded classification results with perfect recall for ArTiSan. However, the cohort contains only two stable eyes, making estimates of discrimination highly uncertain. ArTiSan also shows a substantial increase in AUC relative to GPTS at this threshold. At the -1.0 dB/year threshold, ArTiSan produces improvements in all metrics. At the -1.5 dB/year threshold, ArTiSan improves recall from 0.485 to 0.494, F1-score from 0.546 to 0.564, and AUC from 0.351 to 0.374. Although these results indicate a modest improvement over GPTS at the fixed classification threshold, the low AUC values show that both models have limited ranking

Table 3. Performance comparison between GPTS and ArTiSan across MD-rate thresholds for the LEI-SFD dataset. Bold indicates the better result for each metric and threshold.

Metrics	-0.5 dB/year		-1 dB/year		-1.5 dB/year	
	GPTS	ArTiSan	GPTS	ArTiSan	GPTS	ArTiSan
Accuracy	0.988	0.984	0.756	0.790	0.426	0.457
Precision	0.994	0.994	0.921	0.924	0.626	0.659
Recall	0.994	1.000	0.808	0.844	0.485	0.494
F1-score	0.994	0.997	0.861	0.882	0.546	0.564
AUC	0.623	0.835	0.484	0.527	0.351	0.374

Table 4. Distribution of the highest-weight abnormal archetype among cases misclassified by GPTS but correctly classified by ArTiSan. For each corrected case, the abnormal archetype with the highest weight among A2-A18 was counted once. Values indicate the number of corrected cases associated with each archetype at the -1.0 and -1.5 dB/year thresholds in the UWHVF and LEI-SFD test sets. The final row shows the total frequency across all 72 corrected cases. A1 denotes the normal-field archetype, while A2-A18 denote the abnormal archetypes defined by [10]. A1 was excluded from this analysis, and A2, A3, and A18 were omitted from the table because they were not identified as the highest-weight abnormal archetype in any corrected case. A4-A17 columns are color-coded according to their frequency, with darker colors indicating greater contribution to the corrected cases.

Test set	Threshold	Cases	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15	A16	A17
UWHVF	-1	30	1	0	6	10	1	0	0	2	2	0	2	3	2	1
	-1.5	17	0	0	2	3	1	2	1	2	0	2	2	0	0	2
LEI-SFD	-1	5	0	1	0	0	0	2	0	0	0	1	0	1	0	0
	-1.5	20	1	1	4	3	1	1	0	1	0	2	2	4	0	0
Total corrected cases			2	2	12	16	3	5	1	5	2	5	6	8	2	3

performance on the external cohort. Therefore, the external results should be interpreted as preliminary evidence that archetypal features may improve decisions near the selected operating threshold, rather than as evidence of robust cross-dataset discrimination.

To characterise the cases in which ArTiSan corrected GPTS errors, we identified the highest-weight abnormal archetype for each corrected case at the -1.0 and -1.5 dB/year progression thresholds (Table 4). Of the 72 corrected cases, 52 (72.2%) were assigned to localised VF patterns commonly associated with glaucomatous nerve-fibre damage. These included superior or inferior arcuate defects (A6, A7, and A12; 30 cases, 41.7%), superior or inferior paracentral defects (A8 and A13; 8 cases, 11.1%), superior or inferior nasal-step defects (A10 and A11; 6 cases, 8.3%), and temporal wedge loss (A15; 8 cases, 11.1%). A further 11 cases (15.3%) were associated with superior or inferior altitudinal defects (A9 and A14), which can occur in advanced glaucoma but are less disease-specific. Overall, the findings indicate that ArTiSan-corrected predictions were concentrated in clinically recognisable, glaucoma-compatible VF patterns rather than being distributed arbitrarily across the archetype set.

There are few limitations in this study. First, progression labels were derived from thresholded MD-rate criteria [15], using only 3 VF examinations as input to the model. Alternative global and pointwise trend-based progression definitions have not been explored in this study. Second, the 18 archetypes used in this study include patterns that are not specific to glaucoma, such as hemianopia. Determining the optimal archetype set for progression detection therefore remains an open research question. Finally, external evaluation was limited to one relatively small cohort, with particularly few stable eyes at the -0.5 dB/year threshold, limiting confidence in the corresponding external performance estimates.

Future work should explore anatomically informed tokenisation, such as Garway - Heath sectors [4], as an alternative to fixed 3×3 geometric patching. It would also be valuable to include control experiments that can verify the performance gains stem from clinically meaningful archetypal information. Additionally, analyzing the contribution of individual archetypes could help confirm clinical relevance. Future work should incorporate examination times so that the model captures not only examination order but also irregular inter-visit intervals.

5 Conclusion

Automated glaucoma progression detection from only three baseline visual field (VF) examinations is a challenging learning problem, given the time-intensive nature of longitudinal follow-up. We proposed ArTiSan, an archetype-aware TimeSformer that integrates clinically interpretable pattern-level information via archetypal weight tokens derived from predefined VF loss patterns. The approach improves progression detection over a strong baseline on two datasets, with the largest gains at moderate-to-fast progression thresholds, suggesting that archetypal representations offer useful complementary information for modelling glaucoma progression.

Acknowledgments. This research is supported by Indonesia Endowment Fund for Education (Lembaga Pengelola Dana Pendidikan (LPDP)) scholarships.

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